Experiment 5

Iptables Firewall Configuration

AIM

To configure a packet-filtering firewall using iptables in Kali Linux, implementing rules to allow, deny, and log specific network traffic based on IP addresses, ports, and protocols.

REQUIREMENTS

OS	Kali Linux 2024 (Root Access Required)
Tool	iptables (pre-installed), iptables-save, iptables-restore
Network	Any configured network interface (eth0/lo)
Optional	nmap (for testing firewall rules from another machine)

THEORY

iptables is a user-space utility that allows an administrator to configure the IP packet filter rules of the Linux kernel firewall, using tables, chains, and rules to control network traffic.

Chain	Direction	Description
INPUT	Incoming	Rules for packets destined for the local system
OUTPUT	Outgoing	Rules for packets originating from the local system
FORWARD	Transit	Rules for packets being routed through the system
PREROUTING	Before routing	Modify packets before routing decision (NAT)
POSTROUTING	After routing	Modify packets after routing decision (NAT)

PROCEDURE

- Check current rules: iptables -L -v -n --line-numbers
- Flush all existing rules: iptables -F (to start fresh)
- Set default policies — DROP all INPUT; allow OUTPUT and FORWARD.
- Allow loopback interface traffic (lo) — required for local services.
- Allow established and related connections (for replies).
- Allow incoming SSH on port 22 from a specific IP only.

- Allow incoming HTTP (port 80) and HTTPS (port 443).
- Block a specific IP address from accessing the system.
- Allow ICMP (ping) from the local network only.
- Rate-limit SSH connections to prevent brute-force attacks.
- Log dropped packets with prefix 'IPT-DROP:'.
- Save rules permanently using iptables-save.
- Test rules using nmap from another machine.

COMMANDS

```
# View current rules
iptables -L -v -n --line-numbers

# Flush all rules (clean start)
iptables -F && iptables -X && iptables -Z

# Set default policies
iptables -P INPUT DROP
iptables -P FORWARD DROP
iptables -P OUTPUT ACCEPT

# Allow loopback interface
iptables -A INPUT -i lo -j ACCEPT

# Allow established/related connections
iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT

# Allow SSH from specific IP only
iptables -A INPUT -p tcp --dport 22 -s 192.168.1.100 -j ACCEPT

# Allow HTTP and HTTPS
iptables -A INPUT -p tcp --dport 80 -j ACCEPT
iptables -A INPUT -p tcp --dport 443 -j ACCEPT

# Block specific attacker IP
iptables -A INPUT -s 192.168.1.200 -j DROP

# Allow ping from local network only
iptables -A INPUT -p icmp --icmp-type echo-request -s 192.168.1.0/24 -j ACCEPT

# Rate-limit SSH (max 3 connections per minute)
iptables -A INPUT -p tcp --dport 22 -m limit --limit 3/min -j ACCEPT

# Log dropped packets
iptables -A INPUT -j LOG --log-prefix "IPT-DROP: " --log-level 4

# Save rules to file
iptables-save > /etc/iptables/rules.v4
```

OUTPUT

```
root@kali:~# iptables -L -v -n --line-numbers
Chain INPUT (policy DROP)
num pkts target prot source destination
1 0 ACCEPT all 0.0.0.0/0 0.0.0.0/0 in: lo
2 145 ACCEPT all 0.0.0.0/0 0.0.0.0/0 RELATED,ESTABLISHED
3 0 ACCEPT tcp 192.168.1.100 0.0.0.0/0 dpt:22
4 23 ACCEPT tcp 0.0.0.0/0 0.0.0.0/0 dpt:80
5 5 ACCEPT tcp 0.0.0.0/0 0.0.0.0/0 dpt:443
6 0 DROP all 192.168.1.200 0.0.0.0/0
7 8 ACCEPT icmp 192.168.1.0/24 0.0.0.0/0
8 3 LOG all 0.0.0.0/0 0.0.0.0/0 LOG prefix "IPT-DROP: "
Chain FORWARD (policy DROP)
Chain OUTPUT (policy ACCEPT)

# Nmap scan from another machine (192.168.1.50 -> Kali: 192.168.1.105)
PORT STATE SERVICE
22/tcp filtered ssh [Blocked — not from allowed IP]
80/tcp open http Apache httpd 2.4.57
443/tcp open https Apache httpd 2.4.57
8080/tcp filtered unknown [Dropped by firewall]
3306/tcp filtered mysql [Dropped by firewall]
```

RESULT

iptables firewall was successfully configured with a DROP-by-default policy. SSH access was restricted to a specific IP, HTTP/HTTPS allowed publicly, and the attacker IP (192.168.1.200) was blocked. Nmap scan confirmed that only ports 80 and 443 are accessible. Dropped packets were successfully logged to the kernel log file.